Manthan Sontakke

Nagpur, Maharashtra — +91-8007557800 — manthansontakke005@gmail.com — LinkedIn — GitHub

PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in Artificial Intelligence, Machine Learning, and complex data analysis. Adept at building mathematical models, extracting and manipulating data using SQL, and developing deep learning solutions in Python. Combines strong analytical problem-solving skills with hands-on experience in data visualization to deliver actionable insights.

EDUCATION DETAILS

Vellore Institute of Technology, Bhopal Aug 2023 – May 2027
B.Tech, Electronics and Communication Engineering (AI & Cybernetics) (Full-time) CGPA: 8.66/10

TECHNICAL SKILLS

- **Analytics & Data Management:** Python, SQL, DBMS, Data Extraction, Data Integrity Checks.
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, CNN, NLP, YOLO, Deep Learning, Mathematical Modeling, Digital signal Processing.
- **Data Visualization & Tools:** Streamlit, Plotly, MATLAB, Data Visualization.
- **Cloud & Infrastructure:** AWS (EC2, S3, RDS), Git, Linux, SQLite.
- **Professional Strengths:** Technical Communication, Relationship Building, Team Leadership, Adaptability

PROJECT DETAILS

Cuffless Blood Pressure & Cardiovascular Monitoring System Sep 2026 – Oct 2026
Python, NumPy, SciPy, MAX30102 PPG, MPU-6050 IMU — GitHub

- A non-invasive cardiovascular monitoring system that integrates optical sensing and motion data for continuous vital sign tracking.
- Configured I2C data acquisition scripts for MAX30102 PPG and MPU-6050 IMU sensors, processing hardware sensor data streams utilizing Python, NumPy, and SciPy for signal filtering and analysis.
- Mitigated motion artifacts and extracted reliable physiological features from noisy data streams to estimate blood pressure without traditional inflatable cuffs.

RetailSense AI - Smart Retail Intelligence Platform — GitHub — *Python, YOLOv8, Streamlit, SQLite* Aug 2026

- Developed an AI-powered smart retail intelligence platform that combines computer vision, NLP, and machine learning to track customer activity, detect products, and analyze feedback.
- Engineered the core system architecture utilizing Python, YOLOv8 for real-time product detection, DeepFace for facial recognition, and SQLite, all visualized through an interactive Streamlit dashboard.
- Addressed the lack of real-time retail analytics by automating footfall tracking, product interaction monitoring, and sentiment analysis to improve customer engagement and store operations.

CareVoice - AI-Powered Wellness Check-In Agent — GitHub — *React, Node.js, Web Speech API, SQLite* Mar 2026 – April 2026

- Built an accessible, AI-driven voice check-in system optimized for elderly patients (55+) to continuously monitor daily health metrics, medication adherence, and post-treatment symptoms.
- Developed the full-stack application using React and Node.js, integrating the Web Speech API for voice interactions, Chart.js for data trends, and an explicit rules engine for risk detection.
- Solved the challenge of remote patient monitoring by automatically flagging urgent health risks (e.g., chest pain, missed medications) and synthesizing structured, actionable reports on a centralized clinician dashboard for rapid intervention.

INTERNSHIP EXPERIENCE

Software Intern May 20, 2026 – July 31, 2026
MPOnline Limited

- Developed a Library Management System to manage book records, track book issuance and returns, and organize library resources.
- Implemented core application features and applied programming, debugging, and data management concepts to build a functional software solution.

AI/ML Intern May 20, 2026 – Jul 31, 2026
MPOnline Limited

- Completed a 6-module curriculum covering Python, data preprocessing, and supervised/unsupervised machine learning, applying feature engineering and data visualization techniques to real-world datasets.
- Trained deep learning models using TensorFlow and PyTorch for Computer Vision and NLP tasks across an 11-week program, and deployed an end-to-end AI capstone project via FastAPI following MLOps practices.

CERTIFICATIONS & EXTRA-CURRICULAR

- **Certifications:** Fundamentals of Digital Design for VLSI Chip Design (L&T EduTech / Coursera)
  NPTEL Certified in Internet of Things (Score: 88/100) — Applied Machine Learning in Python (Score: 88.63/100)
- **Extra-Curricular:** Discipline Head, Marathi Cultural Club
  Represented VCA at the State Level Cricket Tournament (Under-17).